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DAY 3

AI IN DAILY LIFE

How Artificial Intelligence Already Runs Your Day

A Course Guide by AYE Tech Hub

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1. Introduction

Most people think Artificial Intelligence is something they will meet *someday* — in a self-driving car, or a futuristic robot. The truth is that AI is already woven into almost every hour of your daily life. From the moment your phone wakes you up to the moment you fall asleep with a podcast playing, dozens of AI systems are quietly making decisions for you and about you.

This lesson — Day 3 of the AYE Tech Hub 30-Day AI Course — shows you exactly where AI lives in your daily routine. You will recognise familiar tools (Google Maps, Netflix, Siri, your bank app) and understand for the first time *how* they work and *why* they behave the way they do. By the end, you will never look at your phone the same way again.

Learning Objectives

By the end of this lesson you will be able to:

- List at least 10 places where AI touches your daily life.
- Explain how recommendation systems decide what to show you.
- Describe the five-step voice assistant pipeline (STT → NLU → action → TTS).
- Identify how Google Maps and Naver Maps choose your route in real time.
- Recognise the role of AI in smart homes, healthcare, banking, and shopping.
- Become a more informed user — understand what data you share, and why.

2. A Day in Your Life — AI Timeline

Let's walk through a typical 24 hours and count every time an AI system touches your life. You will be surprised by how often it happens — most of these touchpoints are invisible because the AI works so smoothly you don't notice it.

A Typical Day — Where You Already Use AI

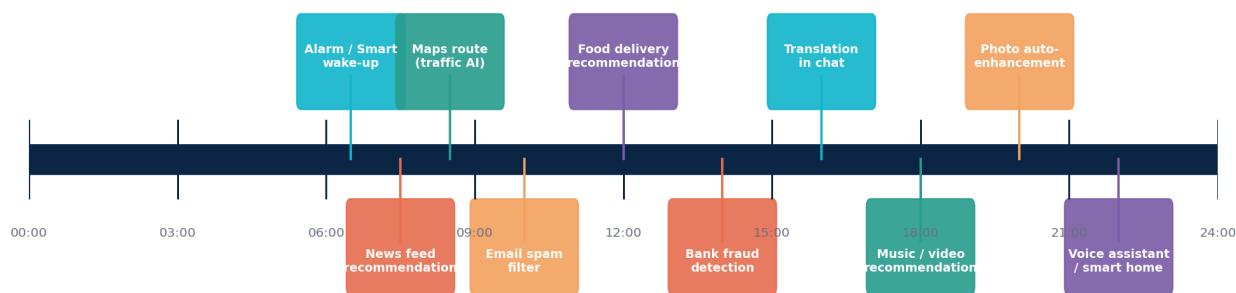


Figure 2.1 — A typical day showing 10 of the most common AI touchpoints. Each coloured block is a separate AI system making a decision for you.

Time	Activity	AI Working Behind the Scenes
06:30	Phone alarm rings	Smart wake-up uses sleep-tracking AI to wake you in a light sleep phase
07:15	Scroll news feed	Personalised ranking AI shows articles based on your past reading
08:00	Drive / commute	Maps AI calculates real-time route using millions of live phone signals
09:30	Read emails	Spam-filter AI hid 30+ junk emails from your inbox automatically
12:00	Order lunch	Delivery app AI recommends restaurants based on your taste history
14:00	Card payment	Fraud-detection AI scans the transaction in milliseconds
16:00	Chat with overseas friend	Translation AI auto-translates messages in 100+ languages
18:30	Watch Netflix / YouTube	Recommendation AI decides what to play next
20:00	Take a photo	Computer vision AI enhances colour, focus, and removes noise
22:00	Ask voice assistant	Speech AI understands your accent and answers your question

Did You Know?

The average smartphone user in Korea touches AI **over 200 times a day** — and most of those interactions take less than a second. You have already been using AI for years; this course will help you understand and use it more intentionally.

3. Recommendation Systems

If you have ever wondered *how Netflix seems to know what you want to watch* — or why YouTube's home page is full of videos that look exactly like the things you watched last week — the answer is **recommendation AI**. This is the single most profitable application of AI today, used by every major content and shopping platform.

How Netflix / YouTube / Spotify Decides What to Show You

AI compares millions of users at once — finds patterns invisible to humans

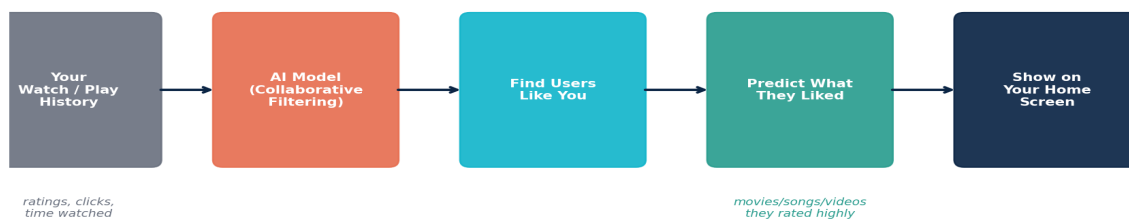


Figure 3.1 — How a recommendation system finds what you'll like. The AI compares you to millions of users with similar tastes and predicts what you'll enjoy next.

How It Actually Works

Recommendation systems use a technique called **collaborative filtering**. The AI builds a mathematical profile of your taste from your behaviour — what you watched, what you skipped, what you re-watched, what you rated. It then finds thousands of other users with similar profiles and looks at *what they liked that you haven't seen yet*. Those titles get pushed to the top of your home screen.

Platform	What It Recommends	Signals It Uses
Netflix	Movies, series, documentaries	Watch time, completion rate, ratings, time of day
YouTube	Videos, Shorts, channels	Watch time, likes, subscriptions, search history
Spotify	Songs, playlists, podcasts	Songs skipped vs played to end, repeat plays, time of day
TikTok	Short videos in 'For You' feed	Watch-to-end rate, likes, shares, re-watches
Amazon / Coupang	Products to buy	Past purchases, items viewed, items others bought together
Instagram / Facebook	Posts, ads, reels	Time spent on each post, likes, comments, shares

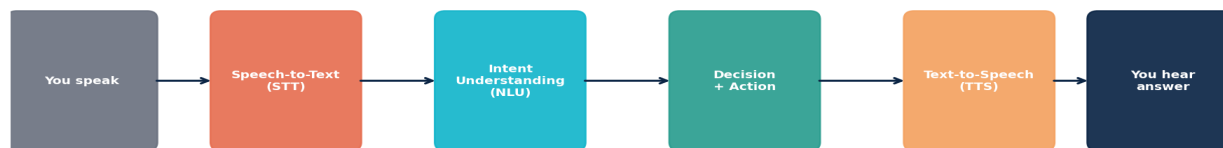
Engineer's Insight

Recommendation AI is what makes streaming apps addictive. The more time you spend, the more data the AI has, and the better its predictions become. This is a feedback loop — and it works in both directions: **great for finding new things you'll love, but also easy to get stuck in a bubble**. Watching one genre for a week trains the AI to show you only that genre. Mix up your viewing if you want broader recommendations.

4. Voice Assistants & Language AI

When you say 'Hey Siri, what's the weather?' or 'Bixby, set a 10-minute timer', your spoken sentence travels through five distinct AI systems in under a second. Each stage solves a different problem — converting sound to text, understanding meaning, deciding what to do, and speaking back to you.

Voice Assistant Pipeline — From Your Words to Action



Example: 'Hey Siri, what's the weather?' → weather API call → 'It's 22°C and sunny.'

Figure 4.1 — The five-stage voice assistant pipeline. Your voice is converted, understood, acted on, and answered — all in roughly half a second.

Stage	AI Technology	What It Does
1. Speech-to-Text (STT)	Acoustic neural network	Converts your audio into written words. Handles accents and noise.
2. Natural Language Understanding (NLU)	Language model	Figures out what you meant. 'Weather?' → request_weather(location)
3. Intent + Action	Rule engine + APIs	Calls the right service (weather API, calendar, music player)
4. Response Generation	Language model	Writes a natural answer: 'It's 22°C and sunny in Gimpo today.'
5. Text-to-Speech (TTS)	Voice synthesis	Speaks the answer back in a human-like voice

Popular Voice Assistants

Assistant	Maker	Strongest Feature
Siri	Apple	Tight iPhone & HomeKit integration
Bixby	Samsung	Deep Samsung phone & SmartThings control
Google Assistant	Google	Best search & translation
Alexa	Amazon	Largest smart-home device ecosystem
Clova	Naver	Best Korean language understanding
ChatGPT Voice	OpenAI	Full-conversation AI, not just commands

Try This

Open your voice assistant and ask the same question three different ways: 'Will it rain tomorrow?' — 'What's the forecast for tomorrow?' — 'Should I bring an umbrella tomorrow?' Notice how the AI understands all three despite the different wording. That is the power of natural language understanding.

5. Maps, Navigation & Ride-Sharing

Every time you open Google Maps, Naver Maps, or Kakao Maps and ask for directions, an AI calculates the best route by combining many live data sources in real time. The road network is a graph with millions of nodes, and the AI must solve a shortest-path problem while accounting for traffic, accidents, and your personal preferences — all in under a second.

How Google Maps / Naver Maps Chooses Your Route

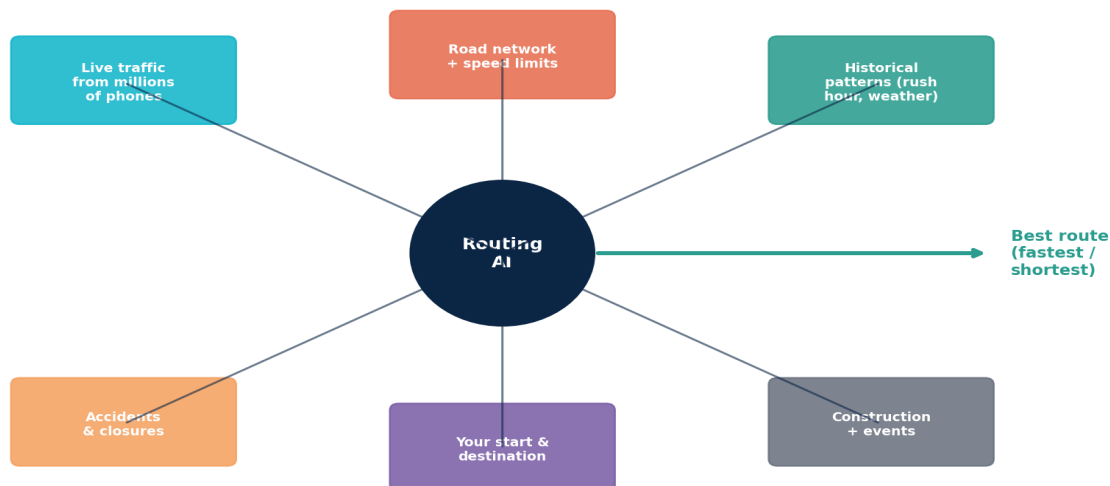


Figure 5.1 — The routing AI fuses six different data streams in real time: live traffic, road network, historical patterns, accidents, your destination, and special events.

How Real-Time Traffic Works

Google and Naver collect anonymous GPS pings from millions of phones running their app. By measuring how fast those phones are moving on each road, the AI builds a live picture of traffic — red on slow roads, green on fast ones. Your phone contributes data too: you are part of the system that gives you the answer.

App	Best For	AI Strength
Google Maps	Worldwide travel	Best global coverage, voice navigation in 50+ languages
Naver Maps	South Korea	Best Korean local data, public transport, indoor maps
Kakao Maps	South Korea	Tight integration with Kakao Taxi and Kakao Pay
Waze	Driving (USA, Europe)	Best community-reported alerts (speed cameras, police)
Uber / Bolt	Ride-sharing	Surge pricing AI, ETA prediction, driver-rider matching
Baemin / Yogiyo	Food delivery (Korea)	Restaurant ranking AI, ETA prediction, courier routing

Did You Know?

When you book a Kakao Taxi or Uber, the AI is solving **two problems at once**: finding the driver who can reach you fastest, AND deciding what to charge based on demand in your area. This is called *dynamic pricing* — and is why prices spike during rush hour or rainy weather. The same algorithm runs in food delivery apps too.

6. Smart Homes & IoT Devices

A modern smart home is a small AI ecosystem. Each connected device — thermostat, camera, speaker, light, lock — collects data, sends it to a cloud AI, and acts on the answer. When done right, the home learns your habits and adapts: lights dim at bedtime, the thermostat warms the bedroom before you wake, the vacuum cleans while you're at work.

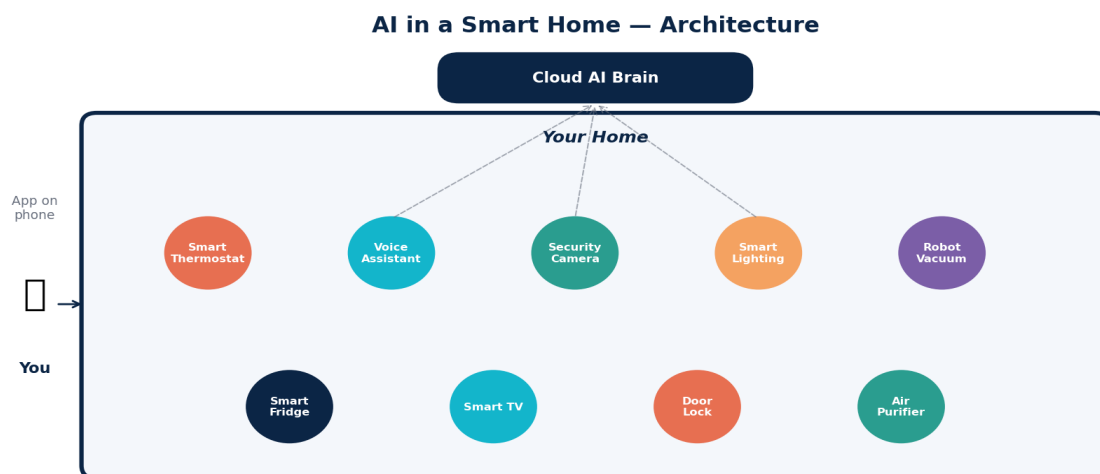


Figure 6.1 — Architecture of a typical AI-enabled smart home. Each device talks to a cloud AI brain that coordinates decisions across the whole house.

Device	What the AI Does	Privacy Consideration
Smart speaker	Listens for wake word; processes commands	Audio sent to cloud after wake word
Security camera	Detects people, packages, animals; sends alerts	Continuous video — review storage settings
Smart thermostat	Learns your schedule; saves 10–20% energy	Occupancy and temperature data uploaded
Robot vacuum	Maps your home; avoids obstacles	Floor plan stored in cloud
Smart lighting	Auto on/off, scenes, ambient sensing	Presence data uploaded
Smart doorbell	Recognises faces, detects packages	Visitor video stored
Smart fridge	Tracks contents, suggests recipes	Camera inside fridge sends images

Privacy First

Smart-home AI works by sending your data to the cloud. That means voice commands, camera footage, and habit data leave your house. Before buying any device, check three things: (1) where the data is stored, (2) who can access it, (3) how to delete it. Reputable brands (Apple, Google, Samsung, LG) publish clear privacy policies — read them.

7. Healthcare & Wearables

AI is moving fastest in personal health. Your smart watch is no longer just a step counter — it is a continuous medical sensor that can detect irregular heart rhythms, falls, low blood oxygen, and even early signs of disease. The data is small, but when an AI looks at millions of similar readings across millions of people, patterns emerge that no individual doctor could see.

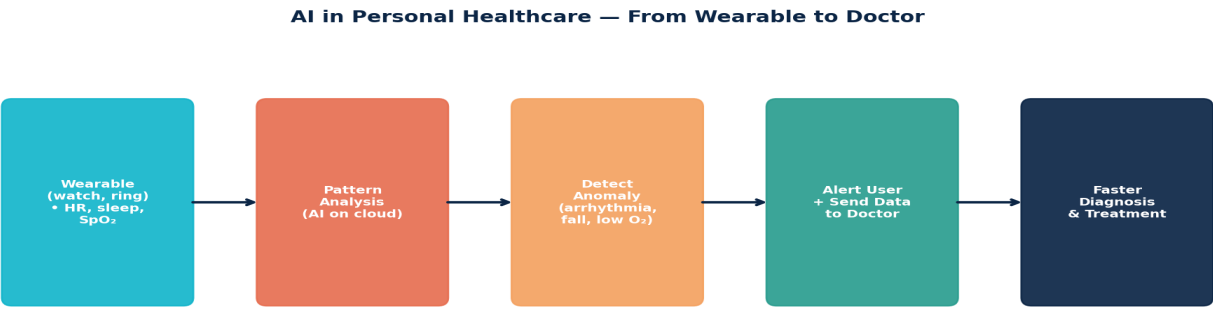


Figure 7.1 — The wearable-to-doctor pipeline. A continuous data stream becomes early warning and faster treatment.

What Your Wearable Can Already Do

Sensor	What It Measures	AI Use
Optical heart rate	Beats per minute, rhythm	Detect atrial fibrillation, abnormal patterns
ECG (Apple Watch, Galaxy Watch)	Electrical heart signal	Hospital-grade rhythm screening
SpO2 pulse oximeter	Blood oxygen saturation	Sleep apnoea, respiratory issues
Accelerometer + gyroscope	Movement, posture	Fall detection, activity tracking
Temperature sensor	Skin temperature trend	Menstrual cycle prediction, illness onset
GPS	Outdoor route, pace	Running form analysis, route safety
Sleep sensor (combined)	Sleep stages, breaks	Sleep quality scoring, smart-alarm wake

Real Example

Apple Watch and Galaxy Watch have already saved thousands of lives by detecting atrial fibrillation (AFib) that wearers did not know they had. The watch detected an irregular rhythm during normal activity, sent an alert, and prompted the user to see a doctor — before they had a stroke. This is AI as a quiet, always-on companion.

8. Finance, Banking & Shopping

Every time you tap your card, transfer money, or buy something online, AI models work in the background — usually in less than 100 milliseconds — to decide whether to approve the transaction. The same AI that approves your normal payment will block a transaction that looks suspicious, even if it's actually you. This is one of the oldest and most valuable applications of AI in daily life.

Use Case	What the AI Decides	How It Helps You
Fraud detection	Is this transaction unusual for this user?	Blocks stolen-card purchases in real time
Credit scoring	How likely is this person to repay a loan?	Decides loan approval and interest rate
Personalised offers	Which promotion would this customer take?	Targeted discounts in banking apps
Chatbot customer service	What is the user asking? Can I answer?	24/7 instant answers without waiting
Stock & FX trading	Should we buy or sell this asset now?	Drives 70%+ of stock-market volume
E-commerce search	Which product matches this query best?	Better search results on Amazon, Coupang

Use Case	What the AI Decides	How It Helps You
Dynamic pricing	What price will this customer pay?	Adjusts hotel, flight, ride prices in real time

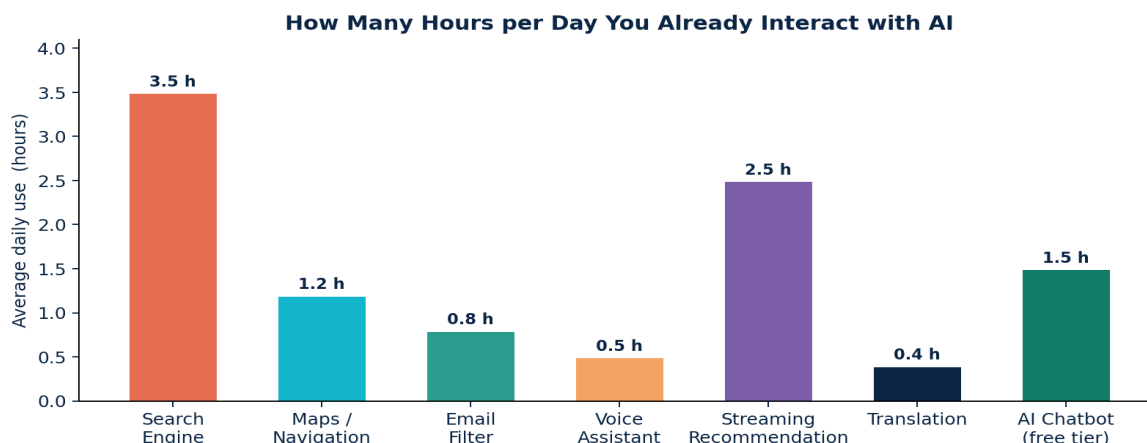


Figure 8.1 — Estimated daily hours an average user spends interacting with AI-driven services. Streaming and search dominate, but background AI (email, payments) runs constantly.

Engineer's Insight

Banks like KB Kookmin, Shinhan, and Hana already use AI to scan every transaction in real time. If your card is suddenly used in a foreign country, the AI compares the transaction to your normal pattern and may block it — even before you notice anything. That is why letting your bank know before travelling is still useful: it adjusts the AI's model of your behaviour.

9. Hidden AI — Behind the Scenes

The most successful AI is the AI you never notice. Below are eight everyday tools that use AI heavily — but never advertise it because the experience is just 'it works'. Recognising these helps you understand how deeply AI is embedded in your daily routine, even when no one calls it AI.

Hidden Use	AI Doing the Work
Email spam filter	Classifies every incoming email as 'inbox' or 'spam' with 99.9% accuracy
Autocorrect & word suggestion	Predicts your next word based on what you typed before
Photo gallery search	Finds photos by content — 'beach', 'dog', 'wedding' — using image recognition
Phone camera enhancement	Combines multiple frames, removes noise, improves dynamic range
Live captions on videos	Real-time speech-to-text in YouTube, Zoom, Teams
Translate (camera mode)	Point the camera, AI reads text and overlays translation
Background blur in video calls	AI segments your body from the background in real time
Password breach alerts	Compares your saved passwords against known breach databases
Search engines (Google, Naver)	Ranks billions of pages to your query in 0.3 seconds
Social media moderation	Auto-detects hate speech, scams, nudity at upload time

Try This Today

Open your phone's photo gallery and search for 'food', 'sunset', or 'document'. Your phone has been quietly tagging every photo with AI labels since the day you bought it — even without internet. This is **on-device AI**, running on the small AI chip inside your phone's processor.

AI Across Industries — A Bigger View

Beyond your phone, AI is reshaping every industry that touches your daily life — even ones you may not see directly. The chart below shows where consumer AI spending is going globally. Recommendation systems and communication tools (translation, chatbots) take more than half the market.

Where Consumer AI Spending Goes (Approximate 2025)

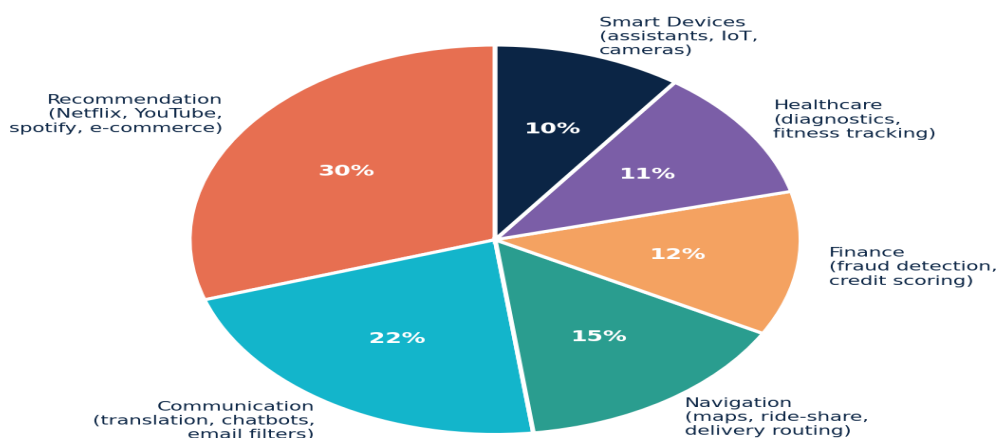


Figure 9.1 — Approximate breakdown of consumer AI spending in 2025 by category. Recommendation systems dominate because they directly drive revenue.

Reflection Question

Think about your last 24 hours. Can you list **at least 10 AI interactions** you had — even ones you didn't notice at the time? Write them down. This single exercise will change how you see your phone, your apps, and your daily routine. AI is not coming — *it is already here, and you are already using it.*

10. Summary, Reflection & Homework

Lesson Recap

- AI is already part of your life at every hour — most touchpoints are invisible because the AI works smoothly.
- **Recommendation systems** drive Netflix, YouTube, Spotify, Amazon — they use collaborative filtering to predict what you'll like.
- **Voice assistants** work in five stages: speech-to-text, natural language understanding, action, response generation, text-to-speech.
- **Maps and navigation** fuse live phone data, road networks, and historical patterns to give you a route in under a second.
- **Smart homes** connect dozens of small AI devices to a cloud brain — convenient, but with privacy implications you should understand.
- **Healthcare wearables** detect heart-rhythm problems, falls, and sleep issues — saving lives every day.
- **Hidden AI** in spam filters, autocorrect, photo enhancement, and search is the most successful kind — you don't notice it because it just works.



Homework Assignment

For the next 24 hours, keep a small AI log:

1. Note every time you notice an AI making a decision for you (recommendation, route, filter, translation, suggestion).
2. Mark which ones **helped you** (saved time, found something good).
3. Mark which ones **distracted or limited you** (kept you scrolling, showed only similar things).
4. Pick one AI tool you currently don't use intentionally — try it tomorrow with one specific goal (e.g., use Google Translate for one full conversation).
5. Write a short paragraph: *How is AI shaping my day, and how do I want to shape that relationship?*

Submit to awetgknway@gmail.com with subject *AI Day 3 — Your Name*.

Coming Up Tomorrow — Day 4

Machine Learning Basics — the engine behind almost everything you saw today. You'll learn the difference between traditional programming and machine learning, the three main types of learning (supervised, unsupervised, reinforcement), and how a computer actually 'learns' from data without being explicitly told what to do.

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